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Lab 2

2a) AverageNumber.java

```java
import java.util.Scanner;

public class AverageNumber {
        public static void main(String[] args) {
                // Welcome message
                System.out.println("This is a program to find average of three number.");

                // Get inputs from user
                Scanner scanner = new Scanner(System.in);
                System.out.println("Enter first number: ");
                double firstNum = scanner.nextDouble();
                System.out.println("Enter second number: ");
                double secondNum = scanner.nextDouble();
                System.out.println("Enter third number: ");
                double thirdNum = scanner.nextDouble();
                scanner.close();

                // Calculate and output the result
                double average = (firstNum + secondNum + thirdNum)/3;
                System.out.println("Average of " + firstNum + ", " + secondNum + ", " + thirdNum + " is " + average);
        }
}
```

2b) DiscountCalculator.java

```java
import java.util.Scanner;

public class DiscountCalculator {
        public static void main(String[] args) {
                // Welcome message
                System.out.println("This is a program to find sale price of an item after a percentage discount.");

                // Get inputs from user
                Scanner scanner = new Scanner(System.in);

                System.out.println("Enter original price of the item (RM): ");
                float originalPrice = scanner.nextFloat();

                // Retype input if percentage discount is not between 0 and 100%
                float discountPercentage;
                while(true) {
                        System.out.println("Enter the percentage discount (%): ");
                        discountPercentage = scanner.nextFloat();
                        if (!(discountPercentage ≥ 0 && discountPercentage ≤ 100)) {
                                System.out.println("Please enter percentage discount between 0% to 100%");
                        }
                        else {
                                break;
                        }
                }
                scanner.close();

                // Calculate and output the result
                double discountedPrice = originalPrice * (100 - discountPercentage) / 100;
                System.out.println(
                                "Price of RM" + String.format("%.2f",originalPrice) +
                                " after getting " + discountPercentage + "% is RM" +
                                String.format("%.2f",discountedPrice)
                                );
        }
}
```

2c) WeightUnitConverter.java

```java
import java.util.Scanner;

public class WeightUnitConverter {
        public static void main(String[] args) {
                // Welcome message
                System.out.println("This is a program to convert weight of parcel in pounds (lbs) to kilograms (kg)");

                // Get inputs from user
                Scanner scanner = new Scanner(System.in);
                System.out.println("Enter weight of parcel in pounds (lbs): ");
                double weightInPounds = scanner.nextDouble();
                scanner.close();

                // Calculate and output the result
                final double convertionFactorFromPoundsToKilograms = 0.45359237;
                double weightInKilograms = weightInPounds * convertionFactorFromPoundsToKilograms;
                System.out.println("Weight of parcel of " + weightInPounds + " lbs is " + String.format("%.6f",weightInKilograms) + " kg");
        }
}
```

2d) BMICalculator.java

```java
import java.util.Scanner;

public class BMICalculator {
        public static void main(String[] args) {
                // Welcome message
                System.out.println("This is a program to calculate Body Mass Index (BMI)");

                // Get inputs from user
                Scanner scanner = new Scanner(System.in);
                System.out.println("Enter weight in kilogram: ");
                double weight = scanner.nextDouble();
                System.out.println("Enter height in meter: ");
                double height = scanner.nextDouble();
                scanner.close();

                // Calculate and output the result
                double bmiValue = weight / (height * height);
                System.out.println("BMI of a person of " + weight + " kg and " + height + " m is " + String.format("%.2f",bmiValue));
        }
}
```